

AI/ML Problem Statements

Problem 1: Mood Prediction from Image–Caption Pairs

Problem

You are given a dataset containing **image–caption pairs**. Each pair consists of:

- A **photo** showing a single person.
- A **caption** describing the context or situation of that photo.
- A **target variable** mood indicating the emotion/feeling experienced by the person in the photo.

Task

Design a complete **pipeline** to predict the mood using both the image and its caption. Your submission should focus on idea and approach (no code or implementation required). Specifically, include:

1. A clear pipeline overview, explicitly listing models and methods you would use.
2. Detailed explanation of the training process: data flow, pretraining / fine-tuning choices, batching, and how multimodal inputs are handled.
3. Loss functions and evaluation metrics (and why each is chosen).
4. Anticipated challenges (e.g., label imbalance, ambiguous captions, domain shift) and how you would mitigate them.
5. A concise diagram of your proposed pipeline.

What we expect

- A well-explained textual overview of your pipeline (you may be as elaborate as you like).
- A figure/diagram illustrating the pipeline (simple block diagram is fine).
- **Note:** We are not asking for code or working implementation — only the idea, approach, and reasoning.

Problem 2: Anti-RAG

Problem

You have access to a **public LLM** (pretrained on large web corpora). Separately, you possess a **confidential document / corpus** whose contents must never be exposed by the public model (neither verbatim nor by paraphrase). The public LLM remains a black box (you may query it), and you may also assume availability of a URL-based retrieval layer (RAG) that can access internet sources.

Task

Design a **robust output-side pipeline** that guarantees responses from the public LLM and retrieval system **do not disclose** or leak information from the confidential corpus. Your pipeline should reduce/limit the model’s “knowledge space” at output time with respect to the confidential corpus.

In your submission (idea/approach only), please include:

1. A clear pipeline overview naming each module/component.
2. How the pipeline prevents retrieval of confidential content (RAG-side controls) and/or prevents the LLM from producing confidential content (output-side controls).
3. Specific methods you would use for redaction, detection, and verification.
4. Training/validation strategy for any detectors or classifiers used (if applicable), including loss functions and evaluation metrics.
5. Failure modes and mitigations (how you detect and handle possible leaks, paraphrase-based disclosures, or hallucinations).
6. A diagram illustrating the Anti-RAG pipeline.

Constraints and assumptions

- The confidential corpus is fixed and known to you (you can compute fingerprints/embeddings/blocklist from it).
- The public LLM is trained on public web data and cannot be edited at the weights-level (you can only control input and output behavior).
- Optionally, you may use URL-based RAG for non-confidential sources; the retriever can be configured to exclude specific URLs or domains (explicitly allowed in more complex designs).

What we expect

- A detailed, well-reasoned pipeline overview (text).
- Brownie points if you submit a figure/diagram of your pipeline to clarify module interactions.
- **Note:** Focus on high-level design, algorithms and rationale — no implementation/code required.

Final note: Both problem statements are intended to evaluate your ability to design principled, defensible AI/ML approaches and to reason about model behavior, risks, and mitigations. Moreover, it would be better if you stick to one detailed approach per P.S. instead of giving multiple solutions. You are encouraged to use the resource sheet and the internet (but not LLMs obvio) to learn and build your approach.